

# The Teammate and Rival Spillover Effect of a Superstar: Evidence from Caitlin Clark and Digital Attention in the WNBA

Applied Econometrics (EC4102) — Mid-Term Research Proposal

Group [Group Number] · [Member Names] · Monsoon Semester, AY 2026–27

## 1 Motivation

In many production settings — firms, research labs, sports teams — a small number of exceptionally productive individuals coexist with many ordinary co-workers. A long-standing question in labor economics is whether such “superstars” generate positive externalities for the people around them, over and above their own output: do co-workers become more visible, more productive, or more valuable simply by association? Answering this matters for how firms structure teams, allocate marketing and promotion budgets around star employees, and negotiate compensation for non-star workers whose market value may be shaped by proximity to the star rather than by their own output alone.

Professional sports offer an unusually clean setting to study this question because output (attention, revenue) is measured daily and at the individual level. The entry of Caitlin Clark into the WNBA in 2024 is a natural case: her presence is associated with well-documented increases in ticket sales, television viewership, and local spending for her own team and for the teams she plays against. What is not known is whether this “superstar effect” extends below the team level — to the individual players who share the court with her, as teammates or as opponents. If it does, the effect has direct relevance to how women's sports leagues and players' associations think about revenue-sharing and individual endorsement value in the current WNBA collective-bargaining debate, and more broadly to the economics of peer and spillover effects in labor markets.

We therefore narrow the broad “superstar externality” question to a precise, measurable one: does sharing the court with a superstar causally raise an individual player's own public attention, net of her own performance that day?

## 2 Literature Survey

The theoretical starting point is **Rosen (1981)**, who shows how small differences in talent can translate into large differences in earnings and visibility once a technology (broadcast, media) allows the most talented performer to be consumed by a very large audience at near-zero marginal cost. This is the mechanism that motivates why a single player like Clark could plausibly move outcomes for an entire league rather than just her own team.

Empirically, **Humphreys and Johnson (2020)** establish the standard design used in this literature: a team–game panel with team and season fixed effects, a star-player indicator, and controls for opponent quality, used to estimate the effect of individual NBA stars on game attendance. This fixed-effects, selection-on-observables approach is the template several recent Clark-specific papers adapt directly. **Stephenson (2024, *Economics Letters*)** uses daily hotel-occupancy data around Iowa City to test whether Clark's collegiate career generated real local economic activity; identification comes from a day-level panel with date and event fixed effects. Notably, the estimated effect shrinks in her senior year relative to her junior year,

which the author attributes to fans substituting from men's to women's games rather than new visitors entering the local economy — a caution that attention-based measures can reflect reallocation rather than net creation of interest.

The most methodologically developed paper to date, “**Superstar Effects in the WNBA**” (2026, *Journal of Sports Economics*), uses a league-wide game-level panel from 1997–2024 covering 27 identified star players and linear/Tobit specifications to show that Clark's home and away attendance effects exceed those of any previous WNBA star, and are larger than published estimates of Michael Jordan's NBA effect. **Zimmer, Snyder and Zimmer (2025, *Economics Bulletin*)** use a similar least-squares panel design to test whether the WNBA's 2024 attendance growth is attributable specifically to the Clark/Reese rookie class rather than to organic, league-wide growth trends.

**The gap.** Every paper above measures a *team- or league-level* outcome — attendance, hotel occupancy, league revenue — and treats Clark's teammates and opponents only as members of the team that benefits. None examine whether the superstar externality is visible at the *individual player* level: does a specific teammate or a specific rival become more publicly visible on days she shares the court with Clark, holding her own on-court performance fixed? This is the standard peer-spillover question from the labor literature (in the spirit of co-worker productivity spillovers), applied here to a new outcome — individual digital public attention, a leading indicator of a player's endorsement and bargaining value — that has not been used in this literature to date.

### 3 The Research Question and the Ideal Experiment

#### 3.1 Statement of the research question

**Main question:** Does sharing the court with Caitlin Clark — as a teammate or as an opposing player — causally increase a WNBA player's public attention, measured by daily Wikipedia pageviews on her own article, holding her own on-court performance that day fixed?

**Sub-question (heterogeneity):** Is the spillover larger for teammates, who share Clark's broadcast slot and locker room, than for rivals, who merely appear opposite her on a given night — and does any effect persist beyond game day, or fully dissipate within 24–48 hours?

#### 3.2 The experimental ideal

**Units (i):** every active WNBA player other than Clark herself, observed on each day of the 2023–2025 seasons (2023 serves as a pre-Clark baseline).

**Treatment ( $D_{it}$ ):** an indicator equal to 1 if player  $i$ 's team plays a game featuring Clark on day  $t$  — either because  $i$  is Clark's teammate (Indiana Fever) or because  $i$ 's team is Clark's opponent that night. In an unconstrained experiment, a researcher would randomly assign, for each player-day, whether that player's game happens to feature Clark, while holding every other determinant of attention — the player's own performance that night, whether the game is nationally televised, opponent strength — fixed by design.

**Outcome ( $Y_{it}$ ):**  $\log(1 + \text{Wikipedia pageviews})$  for player  $i$ 's article on day  $t$  (and, in a secondary specification, day  $t+1$ , to capture next-day attention).

**Identifying comparison:** with random assignment of  $D$ , potential outcomes are independent of treatment,  $(Y^1, Y^0) \perp D$ , so  $E[Y_{it} | D_{it}=1] - E[Y_{it} | D_{it}=0]$  recovers the average causal spillover effect of sharing the court with Clark. The applied challenge is that  $D_{it}$  is not actually randomly assigned — league schedules and broadcast slots are chosen well in advance based on observable team and market characteristics — which motivates the identification strategy in Section 5.

## 4 Prospective Data

**Source and access.** All data are free and already accessible, with no application or paywall:

- Game logs, box scores (points, rebounds, assists, minutes), rosters, and the season broadcast schedule (network/national-TV flag) from Basketball-Reference / Her Hoop Stats.
- Daily, raw (not indexed) pageview counts per player Wikipedia article from the Wikimedia Pageviews REST API ([wikimedia.org/api/rest\\_v1/metrics/pageviews](https://wikimedia.org/api/rest_v1/metrics/pageviews)), available from mid-2015 onward, queryable with no API key.
- Google Trends search-interest indices and Social Blade social-media follower snapshots as supplementary robustness outcomes.

**Structure.** The panel is at the player–day level: roughly 140–150 active players observed across three seasons (one pre-Clark, two with Clark), each season running roughly 120–150 team-game-days, yielding on the order of several thousand player-day observations after restricting to players with an active Wikipedia article throughout the sample window.

**Key variables.** Treatment is split into two dummies,  $\text{TeammateClark}_{it}$  and  $\text{RivalClark}_{it}$ . Controls include the player's own box-score line that day (points, rebounds, assists), a pre-set national-television dummy, home/away status, opponent win percentage entering the game, and the player's pre-sample average pageviews (a baseline popularity measure). **Measurement concerns:** Wikipedia pageview counts include some automated/bot traffic, which the API's *user* access-type filter substantially removes; pageviews can spike for reasons unrelated to Clark (trades, injuries, unrelated news), which we address by controlling for own performance and using player and week fixed effects; and players without an existing Wikipedia article (mostly deep-bench rookies) will be dropped, which we flag as a sample-restriction limitation rather than a random one.

### 4.1 Descriptive statistics

The proposal stage will report, at minimum: mean and standard deviation of daily pageviews on Clark-featuring vs. non-Clark-featuring game days, separately for teammates and rivals; the same comparison for the 2023 pre-Clark season as a placebo check; and an event-time plot of average pageviews in the days surrounding each Clark head-to-head matchup.

## 5 Identification Strategy

### 5.1 Why OLS is not enough

$$Y_{it} = \beta_0 + \beta_1 D_{it} + u_{it}$$

A naive regression of pageviews on the Clark-game indicator is contaminated because  $\text{Cov}(D_{it}, u_{it}) \neq 0$ , for at least two reasons. First, **selection into broadcast slots**: national broadcasters and the league disproportionately schedule marquee, nationally televised games — including many Clark games — involving teams and players who are already more popular for reasons unrelated to Clark (market size, existing star power). This popularity is part of  $u_{it}$  and is correlated with  $D_{it}$ , biasing the OLS estimate of  $\beta_1$  upward. Second, **own-performance endogeneity**: a player may elevate her own play specifically against a marquee opponent (a “play up to the moment” effect), which would itself independently raise attention; without separating this channel from a pure attention spillover, a naive comparison conflates “attention because I played well against a famous opponent” with “attention simply because I was near a famous opponent.” Formally, treated and untreated player-days differ at baseline:  $E[Y_{it}^0 \mid D_{it}=1] \neq E[Y_{it}^0 \mid D_{it}=0]$ , since Clark-game days are disproportionately drawn from already-popular players and marquee broadcast slots.

The confounding path can be sketched as: *Player popularity*  $\rightarrow$  *selection into national-TV slot*  $\rightarrow$  both  $D_{it}$  and  $Y_{it}$ , a backdoor path that a raw comparison does not close.

## 5.2 Empirical strategy

$$Y_{it} = \alpha_i + \lambda_t + \beta_1 \text{TeammateClark}_{it} + \beta_2 \text{RivalClark}_{it} + \beta_3' X_{it} + \varepsilon_{it}$$

where  $\alpha_i$  is a player fixed effect (absorbing each player's time-invariant baseline popularity),  $\lambda_t$  is a week/season fixed effect (absorbing league-wide attention trends, e.g. the general 2024 surge in WNBA interest unrelated to any single game), and  $X_{it}$  is the observable control vector described in Section 4 — own points/rebounds/assists, the national-television dummy, home/away status, and opponent win percentage.

This is a **selection-on-observables** design: conditional on player and time fixed effects and, crucially, on the national-broadcast dummy — which absorbs the main source of selection into high-attention slots — the remaining day-to-day variation in whether a given player's game happens to feature Clark is driven by the season schedule, which is fixed and published months in advance based on prior-season performance and market-rotation rules rather than by that day's realized attention shock. Under this assumption,  $D_{it}$  is plausibly mean-independent of  $\varepsilon_{it}$  conditional on  $\alpha_i$ ,  $\lambda_t$ , and  $X_{it}$ , so  $\beta_1$  and  $\beta_2$  can be interpreted as the causal teammate and rival spillover effects, respectively. We will additionally test  $\beta_1 = \beta_2$  to assess whether teammate proximity generates a larger spillover than mere on-court opposition. As a falsification check, we will re-estimate the model on the 2023 pre-Clark season using a placebo “teammate/rival of [a comparably prominent 2023 star]” indicator, which should show no analogous jump if the design is not simply picking up generic star-adjacency effects.

**Inference (Q4, brief).** The estimating sample is the population of all active WNBA player-days across the three seasons studied (not a survey sample from a larger population), so standard errors are computed to address serial correlation and within-game correlation across teammates rather than sampling uncertainty per se. We will cluster standard errors at the player level as the baseline specification, and report a robustness version clustered at the team-date level to account for the fact that all of a team's players share the same game-day treatment status.

## References

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